Historical References

PDF edition of preserved repository document | docs/historical_references.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Historical external references: suitability audit

These published general-model references are not comparable with our procedural multiplication task. They are contextual records, not achieved Mindscape targets.

Exact checkpoint hashes are not publicly specified in the cited reports; keep them unknown rather than guessing an API snapshot.

Model	Report/version	Dataset/task	Published score	Protocol	Source	Publication date	Suitability
GPT-4	March 2023 technical report evaluation; exact checkpoint undisclosed	GSM8K, grade-school word problems	92.0%	5-shot chain-of-thought; report notes GSM8K training data in pretraining mix	[Table 2 and Appendix E](https://cdn.openai.com/papers/gpt-4.pdf)	Report first submitted 2023-03-15; linked PDF dated 2023-03-27	Not directly comparable
Gemini Ultra	Gemini 1.0, arXiv 2312.11805 v1; exact checkpoint undisclosed	GSM8K, grade-school word problems	94.4%	Chain-of-thought self-consistency, Table 2 labelled Maj1@32; exact shot/temperature not specified in this table	[Technical report, Table 2](https://arxiv.org/html/2312.11805v1)	2023-12-19 submission	Not directly comparable

GPT-4 date metadata: arXiv record (https://arxiv.org/abs/2303.08774).

Gemini date/version metadata: v1 record (https://arxiv.org/abs/2312.11805v1).

Checked against primary reports on 2026-10-08. No live proprietary API was used.

Why a direct claim would be invalid

GSM8K requires understanding word problems and multi-step mathematics. Mindscape's current inputs are two explicit integers with a hand-specified decimal procedure.

The tasks, inputs, training distributions, decoding and compute differ. Published pretraining sizes are not a controlled domain-example budget; a ratio against our 1000 multiplication examples would not be a valid DER. Do not place these scores

on our accuracy/data-efficiency plots or claim that multiplication accuracy matches or beats either system. The two historical protocols also differ from each other.

Methodology for a future genuine comparison

Choose a shared task/dataset/split first. Freeze interpretation, scoring, prompt examples, decoding/sample count, tool access and evaluation budget. Verify that Mindscape actually supports the task; do not carve an easier subset after seeing results. Cite exact published revision/protocol, record any unknown checkpoint or settings, audit contamination and separate reference-only from reproduced results. If exact protocol compatibility cannot be established, report both contextual values with explicit differences and make no superiority or training-data-efficiency claim. Use the internal four-condition controlled comparison as the primary evidence.